



Problem Statement and Engineering Scope

Formal Industry Problem Definition, Operational Context & System Boundaries

Project: AI-Powered Smart Campus Resource Management System	Faculty Coordinator / Mentor: Dr. Rashmi Mothkur (ASAC CSE)
Team Lead: Likith S (2411021061437)	Members: Vinodkumar (2411021061425), Nikhil kumar V (2411021061427), Kalyan kumar T (2411021061421)

1. Official Problem Statement

Problem Statement: Educational institutions often face challenges in efficiently managing campus resources such as classrooms, laboratories, faculty schedules, parking spaces, electricity consumption, and facility maintenance. Resource allocation is typically performed manually, leading to underutilization, scheduling conflicts, increased operational costs, and poor user experience. There is a need for an intelligent campus management platform that can monitor resource utilization in real time, predict future demand, optimize scheduling, reduce energy consumption, and provide data-driven insights for administrators, faculty, and students.

2. Operational Context and Institutional Motivation

At large modern educational institutions such as the 60-acre Alliance University Central Campus in Bengaluru, managing hundreds of instructional spaces across academic blocks (ACED Engineering Block, Central Library, MBA halls, science labs) presents complex operational challenges. Historical data reveals three critical structural inefficiencies:

- **Space Underutilization & Ghost Bookings:** 28.4% of scheduled classroom slots experience ghost bookings—spaces officially reserved on timetables but left unoccupied in practice due to class cancellations or rescheduled sessions, resulting in wasted facility availability.
- **Timetable Clashes & Manual Scheduling Friction:** Manual timetable preparation requires hundreds of staff hours at the start of each semester, frequently resulting in instructor room conflicts, laboratory equipment overlap, and suboptimal student travel intervals.
- **Uncoordinated Energy Consumption & Peak Demand Penalties:** Air conditioning chillers (380–420 kW) and lighting operate on rigid manual schedules rather than dynamic occupancy. Unsynchronized cooling spikes during peak midday hours trigger severe commercial maximum demand surcharges from the regional utility (BESCOM).
- **Parking Congestion & Security Inefficiencies:** Campus vehicular entry and parking allocation operate without real-time occupancy guidance, causing localized congestion during morning student/faculty arrivals.

3. Proposed Solution Scope & System Architecture

The project involves designing and developing a centralized, intelligent Smart Campus Resource Management System. The proposed engineering architecture integrates five core functional tiers:

- **Centralized Multi-Role Management Dashboard:** Responsive web and mobile interfaces tailored for Administrators, Faculty, and Students, enabling seamless facility booking, schedule visualization, parking monitoring, and operational telemetry.
- **Real-Time IoT Telemetry & Occupancy Monitoring:** Ingestion pipeline capturing live data from PIR sensors, thermal probes, smart energy sub-meters, and campus ERP APIs to monitor classroom and laboratory utilization without biometric surveillance.
- **Machine Learning Demand Prediction Core:** Predictive analytics models (LSTM networks, Random Forest, XGBoost) to forecast hourly space demand, identify peak usage periods, and estimate facility maintenance requirements.
- **Automated Timetable & Room Allocation Solver:** Constraint satisfaction algorithms (CSP) and genetic optimization heuristics to automatically generate conflict-free schedules, maximizing seat fill-rate and minimizing room transitions.
- **Energy Consumption Tracking & Recommendation Engine:** Continuous power draw telemetry coupled with automated HVAC setback rules, rooftop solar PV forecasting, and battery storage dispatch recommendations.
- **Integrated Campus Digital Twin:** A lightweight, 3D WebGL cyber-physical digital twin representing campus architecture, enabling intuitive spatial auditing, environmental heatmaps, and closed-loop facility management.

4. Quantitative Baseline vs Engineering Targets

Performance Metric	Current Manual Baseline	Proposed AI Platform Target	Operational Benefit
Classroom Seat Utilization	48.2% average fill-rate	≥ 73.5% optimized fill-rate	+25.3% space efficiency gain
Ghost Booking Wastage	28.4% reserved slots vacant	< 5.0% vacant through auto-release	Reclaims 4.2 hrs/room weekly
Timetable Clashes	12–18 clashes per semester	0 hard clashes (100% elimination)	Zero manual rework overhead
Peak Grid Energy Draw	420 kW uncoordinated peak	≤ 342 kW peak-shaved draw	18.4% reduction in peak load
Facility Booking Latency	24–48 hours manual approval	< 2 seconds instant AI approval	Real-time automated workflow
Administrative Reporting	3–5 days manual aggregation	Real-time 1-click automated export	Continuous audit readiness

5. Curricular Alignment & Course Mapping

In strict compliance with ASAC engineering guidelines, the project synthesizes competencies across 5 mapped courses: **Artificial Intelligence** (Constraint satisfaction timetable solver), **Machine Learning** (LSTM and Random Forest demand forecasting), **Cloud Computing** (AWS/Firebase distributed microservices and database clustering), **Mobile Application Development** (Responsive cross-platform user portals), and **Internet of Things** (PIR occupancy sensors, smart sub-meters, and MQTT edge ingestion).